Table 1: Formal Codebook for Automated Code Review Task Taxonomy (RQ1)

Task & Category	Definition	Inclusion / Exclusion Criteria	Typical Inputs	Typical Outputs	Workflow Example & Representative PSs
Pre-Review Tasks					
Pull Request Assessment	Predicting high-level characteristics, fate, or defect risk of submitted code changes before human inspection.	**Inc:** PR acceptance/abandonment, defect presence, revision necessity. **Exc:** Recommending specific reviewers.	PR metadata, commit churn, diff summary, author history.	Binary label (accept/reject, buggy/clean), review time estimate.	Triages high-risk PRs for senior reviewers before general assignment [P3, P4, P13, P59, P105].
Reviewer Recommendation	Identifying and ranking appropriate developers to inspect a given code change.	**Inc:** Ranking/selecting reviewer IDs based on expertise, file ownership, or social networks. **Exc:** Assigning bug reports outside code review.	Code diff, modified file paths, PR description, reviewer workload.	Ranked list of candidate reviewer IDs, top-k recommendations.	Automatically populates reviewer field on PR creation to reduce assignment latency [P2, P5, P24, P40, P81].
Pull Request Prioritization	Ordering pending pull requests to optimize review queue throughput and urgency.	**Inc:** Scheduling or ranking pending PRs by risk, age, or build status. **Exc:** Within-PR line prioritization.	PR age, test build status, complexity metrics, dependency graph.	Ordered queue of PRs, priority score.	Ranks incoming PRs in team dashboard so urgent or small patches are reviewed first [P23, P37, P49, P116].
Generative Code Review Tasks					
Comment Generation	Automatically synthesizing natural language review comments identifying defects, style issues, or improvements.	**Inc:** Generating feedback sentences for specific diffs/lines. **Exc:** Retrieving verbatim pre-written canned comments.	Code diff, surrounding source code, PR title/description.	Natural language comment text (explanatory or corrective).	AI bot posts inline comments pointing out missing null checks on modified methods [P51, P53, P88, P107, P160].
Code Refinement	Generating revised code patches or fix suggestions that resolve issues raised during review.	**Inc:** Patch generation conditioned on review comments or diffs. **Exc:** Standalone APR without review context.	Old code, code diff, reviewer comment text.	Modified/repaired code snippet (replacement patch).	Developer clicks "Apply AI suggestion" to replace flagged code with refined implementation [P48, P54, P65, P82, P130].
Data Curation	Filtering, denoising, or synthesizing review datasets to train downstream review models.	**Inc:** Automated noise filtering, comment-diff alignment, synthetic review generation. **Exc:** Manual dataset scraping only.	Raw review threads, commit logs, mined diffs.	Cleaned/aligned pairs, quality scores, synthetic training triples.	Pre-filters trivial "LGTM" comments before training generative models [P83, P106, P153, P165].
Assisted Code Review Tasks					

Continued on next page

Table 1 – continued from previous page

Task & Category	Definition	Inclusion / Exclusion Criteria	Typical Inputs	Typical Outputs	Workflow Example & Representative PSs
Issue/Bug Detection	Flagging specific code locations containing potential defects, vulnerabilities, or bad smells.	**Inc:** Line-level or hunk-level vulnerability/bug classification within diffs. **Exc:** Whole-repo static analysis unrelated to review.	Code diff, AST representations, dataflow graphs.	Binary flag (buggy/clean line).	Highlights suspicious array indexing lines in the diff viewer during review [P1, P11, P44, P67, P101].
Comment Recommendation	Retrieving and recommending relevant historical review comments or code snippets for reuse.	**Inc:** Similarity-based retrieval of past discussions/canned feedback. **Exc:** Generating novel comment text from scratch.	Code diff, AST clones, semantic embeddings.	Ranked list of existing historical comments / code snippets.	Suggests standardized feedback previously given on identical clone patterns [P20, P27, P33, P70, P102].
Issue Type Prediction	Categorizing review feedback or changes into taxonomic categories (e.g., bug, style, refactoring).	**Inc:** Multi-class classification of review comments or diff intents. **Exc:** Binary acceptance prediction.	Review comment text, diff hunk, PR tags.	Category label (e.g., Security, Style, Architecture).	Tags review comments automatically to help maintainers route issues to specialists [P8, P14, P52, P78, P139].
Evaluation Tasks					
Comment Evaluation	Assessing the quality, usefulness, actionability, or constructiveness of review comments.	**Inc:** Predicting comment usefulness, actionability, or compliance. **Exc:** Evaluating overall system BLEU scores.	Review comment text, code diff, author response.	Quality score, binary label (actionable/non-actionable).	Flags vague comments ("looks bad") and prompts reviewers to provide actionable feedback [P10, P28, P60, P106, P154].
Sentiment Analysis	Detecting emotional valence, toxicity, or interpersonal civility in review dialogues.	**Inc:** Classifying sentiment (positive/neutral/negative) or toxicity in comments. **Exc:** General social media sentiment analysis.	Review comment text, discussion thread context.	Sentiment polarity, toxicity score, civility label.	Alerts moderators to toxic or unconstructive discussions in pull request threads [P12, P30, P35, P58, P93].
Review Gamification	Modeling incentive and reward mechanisms to encourage thorough and timely review participation.	**Inc:** Automated scoring of reviewer contributions, badges, effort-based incentives. **Exc:** Non-ML leaderboard heuristics.	Reviewer activity logs, comment quality scores, PR turnaround time.	Engagement score, reward tier, gamified feedback.	Computes constructive contribution scores on team review leaderboards [P115].

Task category: ● Pre-Review ● Assisted ● Generative ● Evaluation
Replication package / Dataset availability: ✓ Available ◓ Partially available × Not available

Table 2: Primary studies included in this survey.

ID	Title	Task	Repl.	Dataset
	2015			
[P1]	Supporting Code Review by Automatic Detection of Potentially Buggy Changes	●	✓	×
[P2]	Who should review this change?: Putting text and file location analyses together for more accurate recommendations	●	×	◓
[P3]	Will they like this? evaluating code contributions with language models	●	×	×
	2016			
[P4]	Predicting Defectiveness of Software Patches	●	×	✓
[P5]	Reviewer recommendation for pull-requests in GitHub: What can we learn from code review and bug assignment?	●	×	✓
[P6]	Testing and Code Review Based Effort-Aware Bug Prediction Model	●	×	×
	2017			
[P7]	A case study on machine learning model for code review expert system in software engineering	●	×	×
[P8]	Automatic classification of review comments in pull-based development model	●	×	✓
[P9]	Code review analysis of software system using machine learning techniques	●	×	×
[P10]	Predicting Usefulness of Code Review Comments using Textual Features and Developer Experience	●	✓	✓
[P11]	Vuldigger: A just-in-time and cost-aware tool for digging vulnerability-contributing changes	●	×	×
[P12]	SentiCR: A customized sentiment analysis tool for code review interactions	●	✓	✓
	2018			
[P13]	Early prediction of merged code changes to prioritize reviewing tasks	●	×	×
[P14]	Feedback Topics in Modern Code Review: Automatic Identification and Impact on Changes	●	×	×
[P15]	Intelligent code reviews using deep learning	●	×	×
[P16]	NBSL:A Supervised Classification Model of Pull Request in Github	●	×	×
[P17]	Predicting which pull requests will get reopened in GitHub	●	×	✓
	2019			
[P18]	Accepted or Abandoned? Predicting the Fate of Code Changes	●	×	×
[P19]	Automatic Code Review by Learning the Revision of Source Code	●	×	×
[P20]	Deep Review Sharing	●	×	◓
[P21]	DeepReview: Automatic Code Review Using Deep Multi-Instance Learning	●	×	×
[P22]	Duplicate pull request detection: When time matters	●	×	✓
[P23]	Improving the pull requests review process using learning-to-rank algorithms	●	×	◓
[P24]	Learning to rank reviewers for pull requests	●	×	×
[P25]	Leveraging Change Intents for Characterizing and Identifying Large-Review-Effort Changes	●	×	×
[P26]	Predicting Pull Request Completion Time: A Case Study on Large Scale Cloud Services	●	×	×
[P27]	Style-analyzer: Fixing code style inconsistencies with interpretable unsupervised algorithms	●	✓	✓
[P28]	The Review Linkage Graph for Code Review Analytics	●	✓	✓
	2020			
[P29]	ADCR: An Adaptive TOOL to select Appropriate Developer for Code Review	●	×	✓
[P30]	Affect Recognition in Code Review: An In-situ Biometric Study of Reviewer's Affect	●	×	×
[P31]	Code Inspection Support for Recurring Changes with Deep Learning in Evolving Software	●	×	◓
[P32]	Code review knowledge perception: Fusing multi-features for salient-class location	●	×	×
[P33]	Core: Automating review recommendation for code changes	●	✓	✓
[P34]	CTCPPre: A prediction method for accepted pull requests in GitHub	●	×	×
[P35]	Measuring the effectiveness of code review comments in GitHub repositories: A machine learning approach	●	✓	✓
[P36]	Measuring the Effectiveness of Software Code Review Comments	●	×	×
[P37]	Pull request prioritization algorithm based on acceptance and response probability	●	×	×
[P38]	Recognizing lines of code violating company-specific coding guidelines using machine learning	●	✓	✓
[P39]	Review Sharing via Deep Semi-Supervised Code Clone Detection	●	×	×
[P40]	Using a context-aware approach to recommend code reviewers: findings from an industrial case study	●	×	×
	2021			
[P41]	A Neural Architecture for Detecting Identifier Renaming from Diff	●	✓	×
[P42]	Categorizing Code Review Comments Using Machine Learning	●	×	×
[P43]	Detection and elimination of systematic labeling bias in code reviewer recommendation systems	●	✓	✓
[P44]	GraphEye: A Novel Solution for Detecting Vulnerable Functions Based on Graph Attention Network	●	×	◓
[P45]	iReview: an intelligent code review evaluation tool using biofeedback	●	×	×
[P46]	Predicting design impactful changes in modern code review: A large-scale empirical study	●	✓	✓
[P47]	Recommending pull request reviewers based on code changes	●	×	×
[P48]	Towards Automating Code Review Activities	●	✓	✓
[P49]	Using machine intelligence to prioritise code review requests	●	×	×

2022				
[P50]	Adopting Learning-to-rank Algorithm for Reviewer Recommendation		×	×
[P51]	AUGER: automatically generating review comments with pre-training models		◓	✓
[P52]	Automated Code Review Comment Classification to Improve Modern Code Reviews		✓	✓
[P53]	Automating code review activities by large-scale pre-training		✓	✓
[P54]	AutoTransform: Automated Code Transformation to Support Modern Code Review Process		✓	✓
[P55]	CoDITT5: Pretraining for Source Code and Natural Language Editing		✓	✓
[P56]	Contrastive learning for multi-modal automatic code review		✓	✓
[P57]	DeepLineDP: Towards a Deep Learning Approach for Line-Level Defect Prediction		✓	✓
[P58]	Detecting interpersonal conflict in issues and code review: cross pollinating open-and closed-source approaches		✓	✓
[P59]	Early prediction for merged vs abandoned code changes in modern code reviews		✓	✓
[P60]	Example driven code review explanation		×	×
[P61]	Exploring the notion of risk in code reviewer recommendation		✓	✓
[P62]	Intelligent Code Review Assignment for Large Scale Open Source Software Stacks		×	×
[P63]	Leveraging Test Plan Quality to Improve Code Review Efficacy		×	×
[P64]	Quality Evaluation of Modern Code Reviews Through Intelligent Biometric Program Comprehension		✓	✓
[P65]	Review4Repair: Code review aided automatic program repairing		✓	✓
[P66]	Structuring Meaningful Code Changes in Developer Community		×	×
[P67]	Supporting Code Review by a Neural Network Using Program Images		×	×
[P68]	Turn tree into graph: Automatic code review via simplified ast driven graph convolutional network		✓	✓
[P69]	Using pre-trained models to boost code review automation		✓	✓
[P70]	Where should i look at? recommending lines that reviewers should pay attention to		✓	✓
2023				
[P71]	A Code Reviewer Recommendation Approach Based on Attentive Neighbor Embedding Propagation		×	✓
[P72]	A Flexible Code Review Framework for Combining Defect Detection and Review Comments		×	✓
[P73]	A Multi-Step Learning Approach to Assist Code Review		×	✓
[P74]	Are code review smells and metrics useful in pull request-level software defect prediction?		◓	◓
[P75]	Assessing Code Review Quality with ChatGPT: A Survey of Automated Reviewer Assignment Methods and Experimental Outcomes		×	×
[P76]	Automated Identification of Toxic Code Reviews Using ToxiCR		✓	✓
[P77]	Automatic code review by learning the structure information of code graph		×	×
[P78]	CCT5: A Code-Change-Oriented Pre-trained Model		✓	✓
[P79]	Code Compliance Assessment as a Learning Problem		×	×
[P80]	Code reviewer intelligent prediction in open source industrial software project		×	×
[P81]	Code Reviewer Recommendation for Architecture Violations: An Exploratory Study		✓	✓
[P82]	D-act: Towards diff-aware code transformation for code review under a time-wise evaluation		✓	✓
[P83]	Evacrc: Evaluating code review comments		✓	✓
[P84]	Explaining explanation: An empirical study on explanation in code review		✓	✓
[P85]	Generation-based code review automation: how far are we?		✓	✓
[P86]	Improving Code Refinement for Code Review Via Input Reconstruction and Ensemble Learning		✓	◓
[P87]	Learning to Predict Code Review Completion Time In Modern Code Review		✓	✓
[P88]	LLaMA-Reviewer: Advancing Code Review Automation with Large Language Models through Parameter-Efficient Fine-Tuning		✓	✓
[P89]	Nudge: Accelerating Overdue Pull Requests toward Completion		×	×
[P90]	ReviewRanker: A Semi-supervised Learning-Based Approach for Code Review Quality Estimation		✓	✓
[P91]	Towards Automated Classification of Code Review Feedback to Support Analytics		✓	✓
[P92]	Towards Automated Code Reviews: Does Learning Code Structure Help?		×	◓
[P93]	ToxiSpanSE: An explainable toxicity detection in code review comments		✓	✓
[P94]	Using large-scale heterogeneous graph representation learning for code review recommendations at Microsoft		×	×
2024				
[P95]	A preliminary investigation on using multi-task learning to predict change performance in code reviews		✓	✓
[P96]	ACoRA – A Platform for Automating Code Review Tasks		✓	✓
[P97]	AI-Assisted Assessment of Coding Practices in Modern Code Review		×	×
[P98]	AICodeReview: Advancing code quality with AI-enhanced reviews		✓	×
[P99]	An Empirical Study on Code Review Activity Prediction and Its Impact in Practice		✓	✓
[P100]	An Insight into Security Code Review with LLMs: Capabilities, Obstacles and Influential Factors		✓	✓
[P101]	Automated Code Review and Bug Detection		×	◓
[P102]	Automating modern code review processes with code similarity measurement		✓	✓
[P103]	Code Review Automation Via Multi-task Federated LLM		✓	✓
[P104]	CodeAgent: Autonomous Communicative Agents for Code Review		✓	✓
[P105]	Comparative Study of Reinforcement Learning in GitHub Pull Request Outcome Predictions		✓	✓
[P106]	DeepCRCEval: Revisiting the Evaluation of Code Review Comment Generation		✓	✓
[P107]	Distilling Desired Comments for Enhanced Code Review with Large Language Models		×	◓
[P108]	Distilling Quality Enhancing Comments From Code Reviews to Underpin Reviewer Recommendation		×	×
[P109]	Don't forget to change these functions! recommending co-changed functions in modern code review		✓	✓
[P110]	Exploring the impact of code review factors on the code review comment generation		✓	✓
[P111]	Exploring the potential of ChatGPT in automated code refinement: An empirical study		✓	✓

[P112]	Exploring the Potential of Llama Models in Automated Code Refinement: A Replication Study		✓	✓
[P113]	Fine-tuning and prompt engineering for large language models-based code review automation		✓	✓
[P114]	Fine-Tuning Large Language Models to Improve Accuracy and Comprehensibility of Automated Code Review		×	✓
[P115]	GitRev: An LLM-based Gamification Framework for Modern Code Review Activities		✓	×
[P116]	GPP: A Graph-Powered Prioritizer for Code Review Requests		✓	✓
[P117]	GPT-3.5 for Code Review Automation: How Do Few-Shot Learning, Prompt Design, and Model Fine-Tuning Impact Their Performance?		✓	✓
[P118]	Graph-Based Salient Class Classification in Commits		×	×
[P119]	Improving Automated Code Reviews: Learning from Experience		✓	✓
[P120]	Improving the Learning of Code Review Successive Tasks with Cross-Task Knowledge Distillation		✓	✓
[P121]	Incivility detection in open source code review and issue discussions		✓	✓
[P122]	Knowledge-Guided Prompt Learning for Request Quality Assurance in Public Code Review		✓	◓
[P123]	Leveraging Reviewer Experience in Code Review Comment Generation		✓	✓
[P124]	Named Entity Recognition for Code Review Comments		×	✓
[P125]	On Unified Prompt Tuning for Request Quality Assurance in Public Code Review		×	✓
[P126]	Predicting the First Response Latency of Maintainers and Contributors in Pull Requests		✓	✓
[P127]	Prompting and Fine-tuning Large Language Models for Automated Code Review Comment Generation		✓	✓
[P128]	Resolving Code Review Comments with Machine Learning		×	×
[P129]	RevToken: A Token-Level Review Recommendation: How Far Are We?		✓	✓
[P130]	Structuring Meaningful Code Review Automation in Developer Community		✓	✓
[P131]	Towards Efficient Fine-Tuning of Language Models With Organizational Data for Automated Software Review		×	◓
	2025			
[P132]	A GPT-based Code Review System with Accurate Feedback for Programming Education		×	×
[P133]	A time-aware developer recommendation approach based on multi-feature fusion		×	×
[P134]	AI-Assisted Fixes to Code Review Comments at Scale		×	×
[P135]	Benchmarking and Studying the LLM-based Code Review		×	×
[P136]	Benchmarking LLMs for Fine-Grained Code Review with Enriched Context in Practice		×	×
[P137]	BitsAI-CR: Automated Code Review via LLM in Practice		×	×
[P138]	Breaking Task Isolation: Enhancing Code Review Automation with Mixture-of-Experts Large Language Models		✓	✓
[P139]	Building a Dataset for Combined Classification of Source Code Reviews		×	✓
[P140]	Building Bridges, Not Walls: Fairness-aware and Accurate Recommendation of Code Reviewers via LLM-based Agents Collaboration		✓	✓
[P141]	Code context-based reviewer recommendation		✓	✓
[P142]	Code Review Automation using Retrieval Augmented Generation		✓	✓
[P143]	CodeDoctor: multi-category code review comment generation		✓	✓
[P144]	Combining Large Language Models with Static Analyzers for Code Review Generation		✓	✓
[P145]	CRScore++: Reinforcement Learning with Verifiable Tool and AI Feedback for Code Review		×	✓
[P146]	Divide-and-conquer: Automating code revisions via localization-and-revision		✓	✓
[P147]	E-PRedictor: an approach for early prediction of pull request acceptance		✓	✓
[P148]	Evaluating Large Language Models for Code Review		✓	✓
[P149]	Exploring the Potential of Large Language Models in Fine-Grained Review Comment Classification		✓	✓
[P150]	Fine-Tuning LLMs to Analyze Multiple Dimensions of Code Review: A Maximum Entropy Regulated Long Chain-of-Thought Approach		×	◓
[P151]	Fine-Tuning Multilingual Language Models for Code Review: An Empirical Study on Industrial C# Projects		◓	◓
[P152]	Grounded AI for Code Review: Resource-Efficient Large-Model Serving in Enterprise Pipelines		×	×
[P153]	Harnessing Large Language Models for Curated Code Reviews		✓	✓
[P154]	Hold On! Is My Feedback Useful? Evaluating the Usefulness of Code Review Comments		✓	✓
[P155]	Hydra-Reviewer: A Holistic Multi-Agent System for Automatic Code Review Comment Generation		×	×
[P156]	iCodeReviewer: Improving Secure Code Review with Mixture of Prompts		×	×
[P157]	Impact of LLM-based Review Comment Generation in Practice: A Mixed Open-/Closed-source User Study		✓	✓
[P158]	Intention is All You Need: Refining Your Code from Your Intention		✓	✓
[P159]	Investigating the Understandability of Review Comments on Code Change Requests		✓	✓
[P160]	Issue-Oriented Agent-Based Framework for Automated Review Comment Generation		✓	✓
[P161]	Leveraging Reward Models for Guiding Code Review Comment Generation		✓	✓
[P162]	Prioritizing code review requests to improve review efficiency: a simulation study		×	✓
[P163]	Retrieval-Augmented Code Review Comment Generation		✓	◓
[P164]	SecureReviewer: Enhancing Large Language Models for Secure Code Review through Secure-aware Fine-tuning		✓	✓
[P165]	Too Noisy To Learn: Enhancing Data Quality for Code Review Comment Generation		✓	✓
[P166]	The price of precision: the cost of preprocessing for automated code revision in code review		✓	✓

Primary Studies

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